Lecture 7: Variational Approach to SPDEs

Tien Anh Vu

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Chapter 3: Variational approach to SPDEs

Variational calculus and Dirichlet energy

In ordinary calculus one minimizes a function $j : \mathbb{R}^n \to \mathbb{R}$. In the calculus of variations one minimizes a *functional*

$$J : X \to \mathbb{R}, \qquad v \mapsto J(v),$$

where X is typically an infinite-dimensional function space.

One considers a constrained minimization problem

$$\min_{v \in \mathcal{U} \subset X} J(v),$$

where the admissible set $\mathcal{U}$ encodes constraints such as boundary conditions (e.g. Dirichlet data $v|_{\partial\Omega} = \varphi$). Without such constraints, the Dirichlet-energy problem below has many trivial minimizers (constants).

In the basic Dirichlet-energy example,

$$f^* = \underset{\text{[constraint unclear]}}{\operatorname{argmin}} J, \qquad J(v) := \int_\Omega |\nabla v|^2 \, dx, \qquad \Omega \subset \mathbb{R}^d.$$

This energy penalizes spatial variation: if v changes rapidly then $|\nabla v|$ is large and $J(v)$ is large; if v is constant then $\nabla v = 0$ and $J(v) = 0$.

To ensure $J(v) < \infty$ one typically takes $X = H^1(\Omega)$, the Sobolev space of L^2 functions with L^2 first (weak) derivatives. With Dirichlet boundary constraints, the minimizer f^* is harmonic (in a weak sense).

For an admissible direction g and $0 \le \varepsilon \le 1$,

$$J(f^*) \le J(f^* + \varepsilon g).$$

Here "admissible" means $f^* + \varepsilon g \in \mathcal{U}$ for small ε; for Dirichlet constraints this typically forces $g|_{\partial\Omega} = 0$, i.e. $g \in H^1_0(\Omega)$.

The difference quotient is expanded as

$$\frac{J(f^* + \varepsilon g) - J(f^*)}{\varepsilon} = \frac{1}{\varepsilon} \int_\Omega \Big(|\nabla(f^* + \varepsilon g)|^2 - |\nabla f^*|^2 \Big) \, dx$$

and then (next line)

$$= \frac{1}{\varepsilon} \int_\Omega \left(2\varepsilon \langle \nabla f^*, \nabla g \rangle + \varepsilon^2 |\nabla g|^2 \right) \, dx.$$

Remark 1 (Out-of-source explanation (more explicit algebra)). Using linearity of the gradient,

$$\nabla(f^* + \varepsilon g) = \nabla f^* + \varepsilon \nabla g,$$

and expanding the square,

$$\begin{aligned} |\nabla(f^* + \varepsilon g)|^2 &= |\nabla f^* + \varepsilon \nabla g|^2 \\ &= |\nabla f^*|^2 + 2\varepsilon \langle \nabla f^*, \nabla g \rangle + \varepsilon^2 |\nabla g|^2. \end{aligned}$$

Substituting this into the difference quotient gives

$$\begin{aligned} \frac{J(f^* + \varepsilon g) - J(f^*)}{\varepsilon} &= \frac{1}{\varepsilon} \int_\Omega \Big(2\varepsilon \langle \nabla f^*, \nabla g \rangle + \varepsilon^2 |\nabla g|^2 \Big)\, dx \\ &= 2 \int_\Omega \langle \nabla f^*, \nabla g \rangle \, dx \ + \ \varepsilon \int_\Omega |\nabla g|^2 \, dx. \end{aligned}$$

Sending $\varepsilon \to 0$ yields

$$\frac{d}{d\varepsilon} J(f^* + \varepsilon g) \Big|_{\varepsilon = 0} = 2 \int_\Omega \langle \nabla f^*, \nabla g \rangle \, dx.$$

After a scratched intermediate line, the page records the first-variation condition

$$\int_\Omega \langle \nabla f^*, \nabla g \rangle \, dx = 0.$$

This is the vanishing of the first variation: for $\Phi(\varepsilon) := J(f^* + \varepsilon g)$ one has $\Phi'(0) = 0$ for all admissible g. The identity is already a weak (variational) Euler–Lagrange condition and only involves first derivatives.

Then, by integration by parts (as written),

$$-\int_\Omega \Delta f^* \, g \, dx = 0 \qquad \Rightarrow \qquad \Delta f^* = 0,$$

with the side note "Harmonic function".

Remark 2 (Out-of-source explanation (harmonic function)). A function u is called *harmonic* in Ω if it solves Laplace's equation $\Delta u = 0$ in Ω. Classically this means $u \in C^2(\Omega)$ and $\Delta u(x) = 0$ pointwise. In the variational/weak sense (appropriate for $u \in H^1(\Omega)$), harmonicity means

$$\int_\Omega \nabla u \cdot \nabla \varphi \, dx = 0 \qquad \text{for all } \varphi \in C_c^\infty(\Omega) \text{ (equivalently, for all } \varphi \in H_0^1(\Omega)).$$

Under suitable regularity, the weak and classical notions coincide.

Formally, $\int_\Omega \nabla f^* \cdot \nabla g \, dx = -\int_\Omega (\Delta f^*) \, g \, dx$ plus a boundary term. Under Dirichlet-type variations (where g vanishes on $\partial\Omega$), the boundary term is 0. Interpreted distributionally, this yields $\Delta f^* = 0$ in Ω.

Remark 3 (Out-of-source remark). This is the standard Euler–Lagrange derivation for the Dirichlet energy. With appropriate boundary conditions in the minimization constraint (the subscript is unclear), the minimizer is harmonic in weak form.

Remark 4 (Out-of-source remark (what changes in more general variational problems)). More generally, if one considers a functional of the form

$$J(v) = \int_\Omega L(x, v(x), \nabla v(x))\, dx,$$

then the same variational idea yields the Euler–Lagrange equation

$$\frac{\partial L}{\partial v} - \nabla \cdot \left(\frac{\partial L}{\partial(\nabla v)} \right) = 0,$$

again first in weak form (via testing against admissible directions) and, when v is sufficiently smooth, in strong PDE form. For the Dirichlet energy one has $L(x, v, \nabla v) = |\nabla v|^2$, so $\partial L/\partial v = 0$ and $\partial L/\partial(\nabla v) = 2\nabla v$, giving $-\nabla \cdot (2\nabla v) = 0$, i.e. $\Delta v = 0$.

Hilbert-space setup

Let H be a separable real Hilbert space, and let V be another Hilbert space continuously and densely embedded into H:

$$V \hookrightarrow H.$$

The arrow $V \hookrightarrow H$ indicates an *embedding* $i : V \to H$:

- An *embedding* means that i is *injective*, so we can identify V with its image $i(V) \subset H$ and treat V as a subspace of H.
- In this setting i is assumed *linear* (so $i(\alpha u + \beta v) = \alpha i(u) + \beta i(v)$).
- *Continuously embedded* means i is continuous (bounded): there exists $C > 0$ such that $\|i(u)\|_H \le C\|u\|_V$ for all $u \in V$.
- *Densely embedded* means $i(V)$ is dense in H: every $h \in H$ can be approximated in $\|\cdot\|_H$ by a sequence from $i(V)$ (equivalently, from V after identifying V with its image).
- In many PDE examples one literally has $V \subset H$ as sets of functions and then $i(u) = u$ is just the identity; in the abstract setting one often identifies V with its image $i(V) \subset H$.

Continuity of the embedding means that convergence in V implies convergence in H; equivalently, there exists a constant $C > 0$ such that

$$\|u\|_H \le C\|u\|_V, \qquad \text{for all } u \in V.$$

By rescaling the norm on V (an equivalent norm), one may assume without loss of generality that $C = 1$.

Next identify H with its dual. Denote

$$H' := L(H, \mathbb{R}),$$

the space of continuous linear functionals on H. By the Riesz representation theorem, each $h \in H$ defines a functional $\ell_h \in H'$ via

$$h \longmapsto \ell_h, \qquad \ell_h(g) := \langle h, g\rangle_H, \qquad g \in H,$$

and the map $h \mapsto \ell_h$ is an isometric isomorphism $H \cong H'$.

Remark 5 (Out-of-source explanation (Riesz identification $H \cong H'$)). The statement $H \cong H'$ is the *Riesz representation theorem*: every continuous linear functional $\ell \in H'$ can be written uniquely in the form $\ell = \ell_h$ for some $h \in H$, where

$$\ell_h(g) = \langle h, g \rangle_H \qquad \text{for all } g \in H.$$

Thus H and its dual H' are canonically identified by the inner product on H.

Remark 6 (Out-of-source explanation (why the Riesz map is isometric)). Recall that the dual norm on H' is

$$\|\ell\|_{H'} := \sup_{\|g\|_H \le 1} |\ell(g)|.$$

For $\ell_h(g) = \langle h, g \rangle_H$, Cauchy–Schwarz yields

$$|\ell_h(g)| = |\langle h, g \rangle_H| \le \|h\|_H \, \|g\|_H \le \|h\|_H \qquad (\|g\|_H \le 1),$$

so $\|\ell_h\|_{H'} \le \|h\|_H$. For $h \neq 0$, choose

$$g := \frac{h}{\|h\|_H}, \qquad \text{so that} \qquad \|g\|_H = 1.$$

Then

$$|\ell_h(g)| = |\langle h, g \rangle_H| = \left\langle h, \frac{h}{\|h\|_H} \right\rangle_H = \|h\|_H,$$

which shows $\|\ell_h\|_{H'} \ge \|h\|_H$. Together with the inequality above this yields equality, hence

$$\|\ell_h\|_{H'} = \|h\|_H.$$

In words: it is an *isomorphism* because it is a linear bijection between H and H', and it is *isometric* because it preserves the norm (hence the metric), i.e. $\|\ell_h\|_{H'} = \|h\|_H$.

Remark 7 (Out-of-source remark). This is the standard preliminary setup for the variational approach to SPDEs, where one typically builds the Gelfand triple

$$V \hookrightarrow H \cong H' \hookrightarrow V'.$$

The page appears to be introducing the $V \hookrightarrow H$ embedding and the Riesz identification $H \cong H'$.

Remark 8 (Out-of-source explanation (Gelfand triple)). A *Gelfand triple* (also called a *Hilbert triple*) is a chain of continuous and dense embeddings

$$V \hookrightarrow H \hookrightarrow V',$$

where H is identified with its dual H' via the Riesz map, so one often writes

$$V \hookrightarrow H \cong H' \hookrightarrow V'.$$

This framework is fundamental in the variational approach to PDEs/SPDEs: operators such as the Laplacian are naturally viewed as maps $A : V \to V'$, and solutions are tested against functions in V using the dual pairing ${}_{V'}\langle \cdot, \cdot \rangle_V$.

Embedding $H \cong H' \hookrightarrow V'$

First compute the norm of the Riesz functional:

$$\|\ell_h\|_{H'} \;:=\; \sup_{\|g\|_H \le 1} |\ell_h(g)| \;=\; \|h\|_H.$$

We already proved this above (by Cauchy–Schwarz and the choice $g = h/\|h\|_H$), hence the embedding $H \cong H'$ is an isometry.

In particular,

$$H \cong H' \hookrightarrow V'.$$

The inclusion $H' \hookrightarrow V'$ is the restriction map:

- For $\ell_h \in H'$ (continuous map $H \to \mathbb{R}$), restrict it to $V \subset H$:

$$\ell_h|_V : V \to \mathbb{R}.$$

- Let $(u_n)_n$ be a sequence in V. If $u_n \to u$ in V, then $u_n \to u$ in H, because the embedding $V \hookrightarrow H$ is continuous.
- Since $\ell_h \in H'$ means $\ell_h : H \to \mathbb{R}$ is continuous, it follows that if $u_n \to u$ in H, then $\ell_h(u_n) \to \ell_h(u)$ in $\mathbb{R}$. Hence the restriction $\ell_h|_V$ is continuous on V.
- Therefore $\ell_h \in V'$.

Remark 9 (Out-of-source remark). The map $H' \to V'$ here is the restriction map: every continuous linear functional on H becomes a continuous linear functional on V by restriction, because the embedding $V \hookrightarrow H$ is continuous.

Remark 10 (Out-of-source explanation (why it is called the restriction map)). If $\ell \in H'$ is a functional $\ell : H \to \mathbb{R}$, then its *restriction to* V is the functional $\ell|_V : V \to \mathbb{R}$ defined by

$$(\ell|_V)(v) := \ell(v), \qquad v \in V.$$

This is well-defined because $V \subset H$ (after identifying V with its image under the embedding). Moreover,

$$|(\ell|_V)(v)| = |\ell(v)| \le \|\ell\|_{H'}\, \|v\|_H \le C\, \|\ell\|_{H'}\, \|v\|_V,$$

For fixed $\ell \in H'$, the constant $C\, \|\ell\|_{H'}$ is independent of v, so one may write

$$|(\ell|_V)(v)| \le C_\ell\, \|v\|_V, \qquad C_\ell := C\, \|\ell\|_{H'},$$

which is exactly the boundedness/continuity of $\ell|_V$ on V, i.e. $\ell|_V \in V'$. The map $\ell \mapsto \ell|_V$ is therefore literally "restriction of the domain from H to V".

In summary

$$V \hookrightarrow H \cong H' \hookrightarrow V'.$$

The page writes the induced embedding (via the H-inner product) as

$$v \longmapsto \ell_v : g \longmapsto \big(\langle v, g\rangle_H\big)\big|_V.$$

Domain: H. *Codomain:* V' (each $v \in H$ first defines a functional on H via $h \mapsto \langle v, h\rangle_H$ for $h \in H$, and then we restrict it to V to view it as an element of V', i.e. $g \mapsto \langle v, g\rangle_H$ for $g \in V$).

Then a warning/contrast is written:

$$\text{different from} \quad v \longmapsto \big(g \longmapsto \langle v, g\rangle_V\big).$$

Domain: V. *Codomain:* V' (Riesz map on V induced by $\langle\cdot,\cdot\rangle_V$, sending $v \in V$ to the continuous linear functional on V given by $g \mapsto \langle v, g\rangle_V$).

Next, for the functional ℓ_v on V, the page writes

$$\|\ell_v\|_{V'} \;=\; \max_{\|g\|_V \le 1} \ell_v(g).$$

That is, the operator norm in V' is defined by

$$\|\ell_v\|_{V'} = \sup_{g \in V,\, \|g\|_V \le 1} |\ell_v(g)|.$$

The goal is to show that $\ell_v \in V'$ and to obtain a bound of the form $\|\ell_v\|_{V'} \le C\,\|v\|_H$, i.e. continuity of the canonical embedding $H \hookrightarrow V'$. To estimate this quantity, apply Cauchy–Schwarz in H and then use the continuous embedding $V \hookrightarrow H$:

$$\big|\langle v, g\rangle_H\big| \le \|v\|_H\,\|g\|_H \le\; C\,\|v\|_H\,\|g\|_V \le\; C\,\|v\|_H, \qquad (\|g\|_V \le 1).$$

Taking the supremum over $\|g\|_V \le 1$ yields $\|\ell_v\|_{V'} \le C\,\|v\|_H$, so $\ell_v \in V'$ and the embedding $H \hookrightarrow V'$ is continuous.

Hence:

$$\text{continuity of the embedding } H \hookrightarrow V'.$$

Remark 11 (Out-of-source remark). This is the standard canonical embedding $H \hookrightarrow V'$ associated with the Gelfand triple. The functional assigned to $v \in H$ is $g \mapsto \langle v, g\rangle_H$ for $g \in V$, which is *not* the same construction as the Riesz map on V using $\langle\cdot,\cdot\rangle_V$.

Remark 12 (Out-of-source explanation (what "canonical embedding" means)). In general, a map is called *canonical* if it is determined uniquely by the given structures and does not require extra choices (basis, coordinates, etc.). A *canonical embedding* is a canonical injective map.

Examples:

- (Riesz map) For a Hilbert space H, the map $H \to H'$ given by $h \mapsto \ell_h$ with $\ell_h(g) = \langle h, g\rangle_H$ is canonical.
- (Double dual) For a normed space X, the evaluation map $J : X \to X''$, $(Jx)(\ell) = \ell(x)$ for $\ell \in X'$, is the canonical embedding into the double dual.

In this note, the map $H \to V'$ given by $v \mapsto \ell_v$ with $\ell_v(g) = \langle v, g\rangle_H$ is canonical because it is determined uniquely by the H-inner product and the inclusion $V \hookrightarrow H$. It is an embedding because it is linear and injective, and the estimate $\|\ell_v\|_{V'} \le C\|v\|_H$ shows it is continuous.

Bounded operators $A : V \to V'$ and coercivity

Let

$$A \in L(V, V')$$

be a bounded linear operator $A : V \to V'$.

We will assume that A satisfies a *coercivity condition* with parameters $\lambda \in \mathbb{R}$ and $\alpha > 0$ (see below for a common form of this assumption).

We denote the duality pairing (dualization) between V' and V by ${}_{V'}\langle \cdot, \cdot \rangle_V$. For $g \in V'$ and $u \in V$,

$$ {}_{V'}\langle g, u \rangle_V := g(u). $$

In particular, if $Au \in V'$ and $u \in V$, then ${}_{V'}\langle Au, u \rangle_V$ is well-defined.

Remark 13 (Out-of-source remark). In many SPDE/variational texts, the coercivity assumption for $A \in L(V, V')$ is written in a form such as

$$ {}_{V'}\langle Au, u \rangle_V + \lambda \|u\|_H^2 \geq \alpha \|u\|_V^2, $$

for all $u \in V$. The page appears to be setting up this notation before stating the full inequality.

Intuitively, coercivity means that the "leading part" of A controls the V-norm: the pairing ${}_{V'}\langle Au, u \rangle_V$ grows at least like $\|u\|_V^2$ up to a lower-order correction in the weaker norm $\|u\|_H$.

- *Ellipticity (variational analogue):* the operator is "positive" in the sense that pairing with u produces control of the strong norm: roughly, ${}_{V'}\langle Au, u \rangle_V \gtrsim \|u\|_V^2$ (up to lower-order terms).
- *Dissipation (energy analogue):* in an evolution equation $u' = Au + \cdots$, the same positivity leads to an *energy estimate*: $\|u(t)\|_H^2$ cannot grow too fast and typically satisfies $\frac{1}{2}\frac{d}{dt}\|u(t)\|_H^2 \lesssim -\|u(t)\|_V^2 + \cdots$.

This is the key estimate behind energy bounds and well-posedness.

Coercivity and Sobolev example

The top line continues the dual-pairing example (from the previous page). If $h \in H$, then $\ell_h \in H'$ is defined by $\ell_h(x) = \langle h, x \rangle_H$ for $x \in H$. Restricting ℓ_h to V yields an element of V', and the duality pairing agrees with the H-inner product:

$$ {}_{V'}\langle g, u \rangle_V \;=\; \langle h, u \rangle_H, \qquad g = \ell_h. $$

In other words, ${}_{V'}\langle g, u \rangle_V = g(u)$ extends the notion of an inner product to the situation where the first argument is a dual element $g \in V'$ (which need not lie in H). When g happens to come from some $h \in H$ via the Riesz map, the pairing reduces to $\langle h, u \rangle_H$.

Then the coercivity condition is written explicitly:

$$ \langle Au, u \rangle \geq \alpha \|u\|_V^2 - \lambda \|u\|_H^2, \qquad \forall u \in V. $$

Here $\langle Au, u \rangle$ denotes the dual pairing ${}_{V'}\langle Au, u \rangle_V$.

Next, a coercive-growth intuition is sketched:

$$ J(u) = -\langle Au, u \rangle \longrightarrow \infty \qquad \text{as } \|u\|_V \to +\infty. $$

This expresses coercive growth (often discussed in convex-analysis terms: a coercive, strongly convex functional has a global minimizer). Note that this statement depends on the sign convention for A: if coercivity holds with $\lambda = 0$ as ${}_{V'}\langle Au, u \rangle_V \geq \alpha \|u\|_V^2$, then $-{}_{V'}\langle Au, u \rangle_V \leq -\alpha \|u\|_V^2 \to -\infty$ as $\|u\|_V \to \infty$. For example, if $A = \Delta$ (rather than $A = -\Delta$), then ${}_{V'}\langle \Delta u, u \rangle_V = -\|\nabla u\|_{L^2}^2$ and hence $-\langle \Delta u, u \rangle = \|\nabla u\|_{L^2}^2 \to +\infty$.

Remark 14 (Out-of-source remark). The sketch is illustrating coercive growth: if an energy functional satisfies $J(u) \to +\infty$ as $\|u\|_V \to \infty$, then a minimizing sequence (u_n) with $J(u_n) \downarrow \inf_{u \in V} J(u)$ cannot satisfy $\|u_n\|_V \to \infty$ (otherwise $J(u_n) \to \infty$), so minimizing sequences stay bounded in V. The handwriting informally links this to convex-analysis intuition.

Example: Sobolev space setup

The page then gives the standard PDE example:

$$D \subset \mathbb{R}^d \text{ be an open subset}, \qquad H = L^2(D).$$

Here $H = L^2(D)$ is a Hilbert space with the usual inner product.

$$V := H^1(D) := W^{1,2}(D), \qquad \text{where}$$
$$H^1(D) = \left\{ u \in L^2(D) : \partial_{x_i} u \in L^2(D),\ i = 1, \ldots, d \right\}.$$

In words: $H^1(D)$ consists of functions whose weak first derivatives exist and are square-integrable.

Remark (weak derivative)

The page states a weak-derivative criterion of the form:

$$\exists\, \partial_{x_i} u \in L^2(D) \quad \text{such that} \quad \int_D u\, \partial_{x_i} \varphi\, dx = -\int_D \partial_{x_i} u\, \varphi\, dx \quad \text{for all } \varphi \in C_c^\infty(D).$$

Here $C_c^\infty(D)$ denotes smooth test functions with compact support in D, meaning that $\operatorname{supp}(\varphi) := \overline{\{x \in D : \varphi(x) \neq 0\}}$ is a compact subset of D (equivalently, φ vanishes near ∂D).

Remark 15 (Out-of-source explanation (definition)). We say that u is *weakly differentiable* in the x_i direction if there exists a function $w \in L^1_{\text{loc}}(D)$ (in this note, $w \in L^2(D)$) such that

$$\int_D u\, \partial_{x_i} \varphi\, dx = -\int_D w\, \varphi\, dx \qquad \text{for all } \varphi \in C_c^\infty(D).$$

In that case we define $w := \partial_{x_i} u$ and call it the *weak derivative* of u.

Remark 16 (Out-of-source remark). This is the standard distributional (weak) derivative definition. The identity is an integration-by-parts formula with no boundary term because the test function has compact support in D.

The page ends this visible portion with an "advantage" note:

$$\Rightarrow \text{ Advantage: generalize differentiability.}$$

and an example (in one dimension):

$$u(x) = |x| + 1 \text{ is weakly differentiable}, \qquad u'(x) = \operatorname{sgn}(x).$$

Here the derivative is understood in the weak sense.

Continuation of the example $u(x) = |x| + 1$

One can visualize the sign function together with the note:

Doesn't have to specify at zero; for zero measure set isn't relevant for L^2.

This refers to the value of sgn(0) being irrelevant as an L^2-equivalence-class representative.

Then the weak-derivative check is written for a test function

$$\varphi \in L^1_c((-1,1)).$$

For the integration-by-parts computation below we additionally assume $\varphi \in C^1_c((-1,1))$ (so that φ' exists and φ vanishes near ± 1).

The page computes

$$\int_{-1}^{1} (1+|x|)\,\varphi'(x)\,dx = \int_{-1}^{0} (1-x)\,\varphi'(x)\,dx + \int_{0}^{1} (1+x)\,\varphi'(x)\,dx.$$

After integration by parts on each interval (as written across several lines),

$$\begin{aligned}\int_{-1}^{1} (1+|x|)\,\varphi'(x)\,dx &= \Big[(1-x)\varphi(x)\Big]_{-1}^{0} + \int_{-1}^{0} \varphi(x)\,dx \\ &\quad + \Big[(1+x)\varphi(x)\Big]_{0}^{1} - \int_{0}^{1} \varphi(x)\,dx.\end{aligned}$$

The boundary terms at ± 1 vanish because φ has compact support, and the two boundary terms at 0 cancel.

Hence the result is

$$\int_{-1}^{1} (1+|x|)\,\varphi'(x)\,dx = \int_{-1}^{0} \varphi(x)\,dx - \int_{0}^{1} \varphi(x)\,dx = -\int_{-1}^{1} \operatorname{sgn}(x)\,\varphi(x)\,dx.$$

This matches the weak-derivative convention stated above.

Remark 17 (Out-of-source remark). This verifies that the weak derivative of $u(x) = |x|+1$ is $\operatorname{sgn}(x)$ on $(-1,1)$, with the value at $x=0$ arbitrary (irrelevant in L^2).

Illustration in one dimension: $H^1(0,1)$

In one dimension, a useful classical fact is the following:

$$\text{Illustration } (d=1), \quad D := (0,1), \quad u \in H^1(D)$$

implies that u has an absolutely continuous representative (with respect to Lebesgue measure). Concretely, although u is only defined as an L^2-equivalence class, there exists a representative $\tilde{u} : [0,1] \to \mathbb{R}$ which is absolutely continuous and agrees with u a.e. Moreover, there exists a *density* $u' \in L^2(0,1)$, i.e. the Radon–Nikodym derivative (which coincides a.e. with the weak derivative of u), such that for all $x, y \in (0,1)$,

$$\tilde{u}(x) - \tilde{u}(y) = \int_{y}^{x} u'(s)\,ds,$$

Then:

Remark 18 (Out-of-source remark). In particular, u has a continuous version $\tilde{u}$, which is unique up to equality a.e. (as an L^2 equivalence class). Indeed, if $\tilde{u}_1$ and $\tilde{u}_2$ are continuous representatives with $\tilde{u}_1 = \tilde{u}_2$ a.e., then $w := \tilde{u}_1 - \tilde{u}_2$ is continuous and $w = 0$ a.e., hence $w \equiv 0$ (otherwise continuity would force w to be nonzero on a set of positive measure).

An example representative is indicated:

$$\tilde{u}(x) = \int_{0}^{x} u'(s)\,ds$$

which fixes a particular representative by choosing $\tilde{u}(0) = 0$.

Boundary condition characterization of H_0^1 (1D)

In one dimension, one has the characterization

$$u \in H_0^1(D) \quad \text{iff} \quad u \in H^1(D) \text{ and } \tilde{u}(0) = \tilde{u}(1) = 0.$$

Remark 19 (Out-of-source remark). This is the 1D Dirichlet boundary condition: the (continuous) representative vanishes at the boundary.

$H^1(D)$ as a Hilbert space

Recall that

$$H^1(D) = H^{1,2}(D)$$

is a Hilbert space with inner product

$$\langle u, v \rangle_{H^1(D)} = \int_D u\, v\, dx + \sum_{i=1}^{d} \int_D \partial_{x_i} u\, \partial_{x_i} v\, dx.$$

Norm comparison and Gelfand triple

Equivalently, in vector notation,

$$\langle u, v \rangle_{H^1(D)} = \int_D u\, v\, dx + \int_D \langle \nabla u, \nabla v \rangle_{\mathbb{R}^d}\, dx.$$

In particular,

$$\|u\|^2_{H^1(D)} = \langle u, u \rangle_{H^1(D)} = \int_D |u|^2\, dx + \int_D \langle \nabla u, \nabla u \rangle_{\mathbb{R}^d}\, dx = \int_D |u|^2\, dx + \int_D |\nabla u|^2\, dx.$$

Since $|\nabla u|^2 \geq 0$ pointwise, we have $\int_D |\nabla u|^2\, dx \geq 0$, hence

$$\|u\|^2_{H^1(D)} \geq \int_D |u|^2\, dx = \|u\|^2_{L^2(D)},$$

and taking square roots gives

$$\|u\|_{H^1(D)} \geq \|u\|_{L^2(D)}.$$

This yields the Gelfand triple

$$H^1(D) \hookrightarrow L^2(D) = L^2(D)' \hookrightarrow H^1(D)'.$$

Remark 20 (Out-of-source remark). This is the standard choice for variational PDEs/SPDEs. Depending on boundary conditions, many texts use $V = H_0^1(D)$ instead of $H^1(D)$, leading to $V' = H^{-1}(D)$.

Remark 21 (Out-of-source remark (Sobolev embedding theorem; one example)). On a bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$, the Sobolev embedding theorem says that if $1 \leq p < d$ and

$$p^* := \frac{dp}{d-p},$$

then

$$W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega), \qquad \|u\|_{L^{p^*}(\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)}.$$

For example, in $d = 3$ and $p = 2$ one has $H^1(\Omega) = W^{1,2}(\Omega) \hookrightarrow L^6(\Omega)$.

Weak Laplace operator (variational definition)

Recall the Laplace operator

$$\Delta = \sum_{i=1}^{d} \partial_{x_i}^2.$$

In the variational setting, we define it as a continuous linear operator

$$\Delta : H^1(D) \to H^1(D)'$$

Given $u \in H^1(D)$, define $\Delta u \in H^1(D)'$ by

$$v \longmapsto {}_{H^1(D)'}\langle \Delta u, v \rangle_{H^1(D)} := -\sum_{i=1}^{d} \int_D \partial_{x_i} u \, \partial_{x_i} v \, dx, \qquad v \in H^1(D).$$

This is a consistent extension of the classical Laplacian on smooth functions: if u is smooth enough and v is a smooth test function with compact support in D, then integration by parts yields

$$-\sum_{i=1}^{d} \int_D \partial_{x_i} u \, \partial_{x_i} v \, dx = \int_D (\Delta u) \, v \, dx.$$

Remark 22 (Out-of-source remark). This is the weak/distributional realization of the Laplacian associated with the bilinear form

$$a(u, v) = \int_D \nabla u \cdot \nabla v \, dx.$$

It is the formulation used in finite elements and the variational theory.

A side comment mentions applications/advantages, e.g.:

good for finite element method [and a note about Hilbert-space geometry being easy]

Initial value problem (start of deterministic variational theory)

Consider the abstract evolution equation

$$(*) \quad \begin{cases} \dfrac{d}{dt} u(t) = Au(t) + f(t), & t > 0, \\ u(0) = u_0. \end{cases}$$

This includes, for example, the inhomogeneous heat equation when $A = \Delta$.

Theorem 3.2 (Lions variational well-posedness)

Theorem 1 (Standard variational well-posedness)**.** Assume $A \in L(V, V')$ is coercive. Let $u_0 \in H$ and $f \in L^2([0, T]; V')$. Then there exists a unique solution u such that

$$u \in L^2([0, T]; V) \cap C([0, T]; H), \qquad u' \in L^2([0, T]; V'),$$

and u satisfies $u'(t) = Au(t) + f(t)$ in V' for a.e. $t \in (0, T)$ with $u(0) = u_0$.

Remark 23 (Out-of-source remark). This is the standard Lions variational well-posedness theorem for deterministic evolution equations in a Gelfand triple. A common full statement concludes

$$u' \in L^2([0,T];V'), \qquad u \in L^2([0,T];V) \cap C([0,T];H).$$

Remark 24 (Out-of-source explanation (why these spaces appear)). The condition $u \in L^2([0,T];V)$ ensures that $u(t) \in V$ for a.e. t, so $Au(t)$ is well-defined as an element of V' and the natural "energy" $\int_0^T \|u(t)\|_V^2\,dt$ is finite. The condition $u' \in L^2([0,T];V')$ means the time derivative exists in the weak sense with values in V', so the equation $u'(t) = Au(t) + f(t)$ is meaningful in V'. Finally, $u \in C([0,T];H)$ allows one to interpret the initial condition $u(0) = u_0 \in H$ and to evaluate $\|u(t)\|_H$ pointwise in time; in fact, $u \in C([0,T];H)$ typically follows from $u \in L^2([0,T];V)$ and $u' \in L^2([0,T];V')$ via the energy identity below.

Auxiliary lemma: energy identity in the Gelfand triple

Lemma 1 (Energy identity)**.** Let $u \in L^2([0,T];V)$ with $u' \in L^2([0,T];V')$. Then $u \in C([0,T];H)$ and

$$\frac{1}{2}\frac{d}{dt}\|u(t)\|_H^2 = {}_{V'}\left\langle u'(t), u(t)\right\rangle_V \quad \text{for a.e. } t \in (0,T).$$

Proof (as written: Step 1)

Choose $u_m \in C^1([0,T];V)$ such that

$$\lim_{m\to\infty}\Big(\|u_m - u\|^2_{L^2([0,T];V)} + \|\dot{u}_m - \dot{u}\|^2_{L^2([0,T];V')}\Big) = 0.$$

In particular, this assumption means that $u_m \to u$ *strongly* in $L^2([0,T];V)$ and $\dot{u}_m \to \dot{u}$ *strongly* in $L^2([0,T];V')$.

Then

$$\frac{d}{dt}\|u_m(t)\|_H^2 = 2\left\langle \frac{du_m}{dt}(t), u_m(t)\right\rangle_H = 2\,{}_{V'}\left\langle \frac{du_m}{dt}(t), u_m(t)\right\rangle_V.$$

The first equality is the usual chain rule in a real Hilbert space: since $\|u_m(t)\|_H^2 = \langle u_m(t), u_m(t)\rangle_H$ and u_m is C^1 in time,

$$\frac{d}{dt}\langle u_m(t), u_m(t)\rangle_H = \langle \dot{u}_m(t), u_m(t)\rangle_H + \langle u_m(t), \dot{u}_m(t)\rangle_H = 2\,\langle \dot{u}_m(t), u_m(t)\rangle_H.$$

The second equality uses the Gelfand triple: $u_m(t) \in V$ and $\dot{u}_m(t) \in V \subset H \hookrightarrow V'$, and for elements of H the duality pairing agrees with the H-inner product, i.e. ${}_{V'}\langle \dot{u}_m(t), u_m(t)\rangle_V = \langle \dot{u}_m(t), u_m(t)\rangle_H$. Note that $u_m(t) \in V$ (and $\dot{u}_m(t) \in V$) holds for every $t \in [0,T]$ because $u_m \in C^1([0,T];V)$ by construction.

Passing $m \to \infty$, the page records

$$2\,{}_{V'}\left\langle \frac{du_m}{dt}(t), u_m(t)\right\rangle_V \longrightarrow 2\,{}_{V'}\left\langle \frac{du}{dt}(t), u(t)\right\rangle_V \quad \text{in } L^1([0,T]),$$

which follows from Cauchy–Schwarz/Hölder ($p = q = 2$) and the strong convergence in $L^2([0,T];V)$ and $L^2([0,T];V')$. In particular, the L^1 convergence can be checked by estimating the L^1-norm of the difference (see below); equivalently, after extracting a subsequence with a.e. convergence,

one may justify passing $m \to \infty$ under the time integral using Lebesgue's convergence theorem (dominated convergence). Indeed, write

$$ {}_{V'}\langle \dot{u}_m, u_m\rangle_V - {}_{V'}\langle \dot{u}, u\rangle_V = {}_{V'}\langle \dot{u}_m - \dot{u},\, u_m\rangle_V + {}_{V'}\langle \dot{u},\, u_m - u\rangle_V \,. $$

For each $t \in [0,T]$, apply the duality estimate $|{}_{V'}\langle a, b\rangle_V| \le \|a\|_{V'}\|b\|_V$ to both terms:

$$ |{}_{V'}\langle \dot{u}_m(t) - \dot{u}(t),\, u_m(t)\rangle_V| \le \|\dot{u}_m(t) - \dot{u}(t)\|_{V'}\, \|u_m(t)\|_V, $$

$$ |{}_{V'}\langle \dot{u}(t),\, u_m(t) - u(t)\rangle_V| \le \|\dot{u}(t)\|_{V'}\, \|u_m(t) - u(t)\|_V. $$

Now integrate over $t \in [0,T]$ and use Cauchy–Schwarz in time:

$$ \begin{aligned} \|{}_{V'}\langle \dot{u}_m, u_m\rangle_V - {}_{V'}\langle \dot{u}, u\rangle_V\|_{L^1([0,T])} &= \int_0^T |{}_{V'}\langle \dot{u}_m(t), u_m(t)\rangle_V - {}_{V'}\langle \dot{u}(t), u(t)\rangle_V|\, dt \\ &= \int_0^T |{}_{V'}\langle \dot{u}_m(t) - \dot{u}(t), u_m(t)\rangle_V + {}_{V'}\langle \dot{u}(t), u_m(t) - u(t)\rangle_V|\, dt \\ &\le \int_0^T |{}_{V'}\langle \dot{u}_m(t) - \dot{u}(t), u_m(t)\rangle_V|\, dt + \int_0^T |{}_{V'}\langle \dot{u}(t), u_m(t) - u(t)\rangle_V|\, dt \\ &\le \int_0^T \|\dot{u}_m(t) - \dot{u}(t)\|_{V'}\, \|u_m(t)\|_V\, dt + \int_0^T \|\dot{u}(t)\|_{V'}\, \|u_m(t) - u(t)\|_V\, dt. \end{aligned} $$

Finally, apply Cauchy–Schwarz to each integral (Hölder's inequality with $p = q = 2$):

$$ \begin{aligned} \int_0^T \|\dot{u}_m(t) - \dot{u}(t)\|_{V'}\, \|u_m(t)\|_V\, dt &\le \left(\int_0^T \|\dot{u}_m(t) - \dot{u}(t)\|_{V'}^2\, dt\right)^{1/2} \left(\int_0^T \|u_m(t)\|_V^2\, dt\right)^{1/2} \\ &= \|\dot{u}_m - \dot{u}\|_{L^2([0,T];V')}\, \|u_m\|_{L^2([0,T];V)}, \\ \int_0^T \|\dot{u}(t)\|_{V'}\, \|u_m(t) - u(t)\|_V\, dt &\le \left(\int_0^T \|\dot{u}(t)\|_{V'}^2\, dt\right)^{1/2} \left(\int_0^T \|u_m(t) - u(t)\|_V^2\, dt\right)^{1/2} \\ &= \|\dot{u}\|_{L^2([0,T];V')}\, \|u_m - u\|_{L^2([0,T];V)}. \end{aligned} $$

Combining these inequalities gives

$$ \|{}_{V'}\langle \dot{u}_m, u_m\rangle_V - {}_{V'}\langle \dot{u}, u\rangle_V\|_{L^1([0,T])} \le \|\dot{u}_m - \dot{u}\|_{L^2([0,T];V')}\, \|u_m\|_{L^2([0,T];V)} + \|\dot{u}\|_{L^2([0,T];V')}\, \|u_m - u\|_{L^2([0,T];V)}. $$

Since $\|u_m\|_{L^2([0,T];V)} \le \|u_m - u\|_{L^2([0,T];V)} + \|u\|_{L^2([0,T];V)}$, the right-hand side tends to 0 by the assumed strong convergences. Therefore we have proved

$$ 2\, {}_{V'}\left\langle \frac{du_m}{dt}(t), u_m(t)\right\rangle_V \longrightarrow 2\, {}_{V'}\left\langle \frac{du}{dt}(t), u(t)\right\rangle_V \quad \text{in } L^1([0,T]). $$

Also,

$$ \lim_{m\to\infty} \|u_m(t)\|_H^2 = \|u(t)\|_H^2 \quad \text{in } L^1([0,T]). $$

Hence it follows that

$$ t \mapsto \|u(t)\|_H^2 $$

is absolutely continuous with derivative

$$ 2\, {}_{V'}\left\langle \frac{du}{dt}(t), u(t)\right\rangle_V . $$

Here "absolutely continuous" is meant in the standard real-variable sense: a function $f : [0,T] \to \mathbb{R}$ is absolutely continuous if it satisfies either (equivalently both) of the following.

- (ε–δ definition) For every $\varepsilon > 0$ there exists $\delta > 0$ such that for any finite family of pairwise disjoint intervals $(a_k, b_k) \subset [0, T]$,

$$\sum_k (b_k - a_k) < \delta \quad \Longrightarrow \quad \sum_k |f(b_k) - f(a_k)| < \varepsilon.$$

- (Integral/derivative characterization) There exists $g \in L^1([0, T])$ such that

$$f(t) = f(0) + \int_0^t g(s)\, ds \qquad \text{for all } t \in [0, T].$$

  In that case f is differentiable for a.e. t and $f'(t) = g(t)$ a.e.

Remark 25 (Out-of-source remark). This is the standard Hilbert-space chain rule for $u \in L^2([0, T]; V)$ with $u' \in L^2([0, T]; V')$. In many texts this is first proved for smooth approximations and then passed to the limit exactly as sketched here.

Proof (as written: Step 2)

Define the time-average:

$$\langle v \rangle := \frac{1}{T} \int_0^T v(t)\, dt, \qquad v \in L^2([0, T]; H).$$

Then for $v \in C^1([0, T]; V)$,

$$\|v(t)\|_H^2 = \|v(t) - \langle v \rangle + \langle v \rangle\|_H^2 \le 2\|v(t) - \langle v \rangle\|_H^2 + 2\|\langle v \rangle\|_H^2.$$

Here we used the elementary inequality $\|a + b\|_H^2 \le 2\|a\|_H^2 + 2\|b\|_H^2$ (take $a = v(t) - \langle v \rangle$ and $b = \langle v \rangle$). The time average $\langle v \rangle = \frac{1}{T} \int_0^T v(s)\, ds$ is well-defined in H since $v \in L^2([0, T]; H)$ implies $v \in L^1([0, T]; H)$ on a finite interval.

Using

$$v(t) - \langle v \rangle = \frac{1}{T} \int_0^T \big(v(t) - v(r)\big)\, dr,$$

We estimate $\|v(t) - \langle v \rangle\|_H^2$ by Jensen's inequality. Recall: if $\Phi : \mathbb{R} \to \mathbb{R}$ is convex and μ is a probability measure, then

$$\Phi\left(\int f\, d\mu\right) \le \int \Phi(f)\, d\mu.$$

Here f is an integrable function on the underlying measure space (i.e. $f \in L^1(\mu)$). Here we apply this with the probability measure $\mu = \frac{1}{T}\, dr$ on $[0, T]$ and the convex function $\Phi(x) = \|x\|_H^2$ (convex on the Hilbert space H). In this application, f is the H-valued function $f(r) := v(t) - v(r)$ (with t fixed).

$$\left\| \frac{1}{T} \int_0^T \big(v(t) - v(r)\big)\, dr \right\|_H^2 \le \frac{1}{T} \int_0^T \|v(t) - v(r)\|_H^2\, dr.$$

Similarly, for the average term one has (again by Jensen)

$$\|\langle v \rangle\|_H^2 = \left\| \frac{1}{T} \int_0^T v(r)\, dr \right\|_H^2 \le \frac{1}{T} \int_0^T \|v(r)\|_H^2\, dr.$$

Combining these bounds gives

$$\|v(t)\|_H^2 \le \frac{2}{T}\int_0^T \|v(t)-v(r)\|_H^2\,dr + \frac{2}{T}\int_0^T \|v(r)\|_H^2\,dr.$$

For the first term, assume $t \ge r$:

$$\|v(t)-v(r)\|_H^2 = \int_r^t \frac{d}{ds}\|v(s)-v(r)\|_H^2\,ds$$

This is just the fundamental theorem of calculus applied to the real-valued function $F(s) := \|v(s)-v(r)\|_H^2$ on $[r,t]$, using $F(r)=0$.

$$= 2\int_r^t {}_{V'}\left\langle \frac{dv}{ds}(s),\, v(s)-v(r)\right\rangle_V ds$$

$$\le 2\int_r^t \left\|\frac{dv}{ds}(s)\right\|_{V'} \|v(s)-v(r)\|_V\,ds$$

$$\le \int_r^t \left\|\frac{dv}{ds}(s)\right\|_{V'}^2 ds + \int_r^t \|v(s)-v(r)\|_V^2\,ds.$$

Substituting this estimate into the previous inequality gives a bound of the form

$$\|v(t)\|_H^2 \le \frac{2}{T}\int_0^T \left\|\frac{dv}{ds}(s)\right\|_{V'}^2 ds + \frac{2}{T}\int_0^T\int_0^T \|v(s)-v(r)\|_V^2\,ds\,dr + \frac{2}{T}\int_0^T \|v(r)\|_H^2\,dr.$$

More explicitly: starting from

$$\|v(t)\|_H^2 \le \frac{2}{T}\int_0^T \|v(t)-v(r)\|_H^2\,dr + \frac{2}{T}\int_0^T \|v(r)\|_H^2\,dr,$$

and using for each $r \in [0,T]$ the estimate (valid e.g. when $t \ge r$)

$$\|v(t)-v(r)\|_H^2 \le \int_r^t \left\|\frac{dv}{ds}(s)\right\|_{V'}^2 ds + \int_r^t \|v(s)-v(r)\|_V^2\,ds,$$

we obtain

$$\frac{2}{T}\int_0^T \|v(t)-v(r)\|_H^2\,dr \le \frac{2}{T}\int_0^T\int_r^t \left\|\frac{dv}{ds}(s)\right\|_{V'}^2 ds\,dr + \frac{2}{T}\int_0^T\int_r^t \|v(s)-v(r)\|_V^2\,ds\,dr.$$

By enlarging the integration domains (so that $\int_r^t \le \int_0^T$) one can bound

$$\int_0^T\int_r^t \left\|\frac{dv}{ds}(s)\right\|_{V'}^2 ds\,dr \le \int_0^T\int_0^T \left\|\frac{dv}{ds}(s)\right\|_{V'}^2 ds\,dr = T\int_0^T \left\|\frac{dv}{ds}(s)\right\|_{V'}^2 ds,$$

and similarly

$$\int_0^T\int_r^t \|v(s)-v(r)\|_V^2\,ds\,dr \le \int_0^T\int_0^T \|v(s)-v(r)\|_V^2\,ds\,dr.$$

Substituting these bounds yields the displayed estimate. The purpose of this calculation is to obtain a pointwise-in-time bound on $\|v(t)\|_H^2$ in terms of time-integrated quantities. In particular, one obtains an estimate of the form

$$\sup_{t\in[0,T]} \|v(t)\|_H^2 \;\le\; C\left(\left\|\frac{dv}{dt}\right\|^2_{L^2([0,T];V')} + \|v\|^2_{L^2([0,T];V)} + \|v\|^2_{L^2([0,T];H)}\right),$$

for a constant C depending only on T (and the embedding constants). This is the "desired form" used below. Substituting $v = u_n - u_m$ then shows that if (u_n) is Cauchy in $L^2([0,T];V)$ and $(\dot{u}_n)$ is Cauchy in $L^2([0,T];V')$, it is also Cauchy in $C([0,T];H)$. Since H is complete, $C([0,T];H)$ is complete under the supremum norm $\|w\|_{C([0,T];H)} := \sup_{t\in[0,T]} \|w(t)\|_H$. Hence (u_n) converges to some $u \in C([0,T];H)$.

Continuation / closing lines

Then a short "Final remark" is written:

$$u \in H^1([0,T]), \qquad u(y) - u(x) = \int_x^y u'(s)\,ds.$$

Remark 26 (Out-of-source remark). This final line is the 1D Sobolev/absolute-continuity identity in time, matching the idea that H^1 functions have absolutely continuous representatives.